

Mathematical Modelling of Stick-Slip Vibrations: From Belt-Driven Model to Bowed String Instruments

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Introduction:

Stick-slip motion arises whenever two surfaces in contact alternate between periods of static adhesion and kinetic sliding. This cyclical behaviour occurs across a range of scales, from the motion of tectonic plates that produces earthquakes, to the squeal of brake pads against a disc, to the everyday experience of a door hinge creaking. The common theme is a difference between static and dynamic friction: a surface "sticks" until the driving force exceeds the static friction threshold, then "slips" as the lower dynamic friction takes over, before eventually coming back into contact and sticking again.

One of the most musically significant examples of stick-slip motion is the interaction between a bow and a vibrating string. When a violinist or cellist draws a bow across a string, rosin on the bow hair creates a high friction contact. The string initially sticks to the bow and is dragged along with it, building up elastic restoring force. Once the tension exceeds the static friction limit, the string breaks free and slips back, oscillating under the influence of its own elasticity and the lower kinetic friction. It then recaptures on the bow and the cycle repeats. This periodic stick-slip process is what produces a sustained, controllable pitch, the so-called Helmholtz motion that is fundamental to the tone of all bowed string instruments. A similar mechanism is at play when a guitarist produces a harmonic squeal by lightly dragging a finger along a wound string, or when a wet finger is rubbed around the rim of a wine glass to produce a singing tone.

To investigate the essential physics of this behaviour, this report develops a simplified mathematical model based on a mass–spring system resting on a moving belt. The block is attached to a fixed wall by a spring of stiffness k and sits on a belt moving at constant velocity v . Friction between the block and belt is characterised by a static coefficient μ_s and a dynamic coefficient μ_d , with $\mu_s > \mu_d$. During the stick phase the block moves with the belt and the spring stretches; during the slip phase the block oscillates under the competing influence of the spring restoring force and dynamic friction. This idealised setup captures the main mechanism.

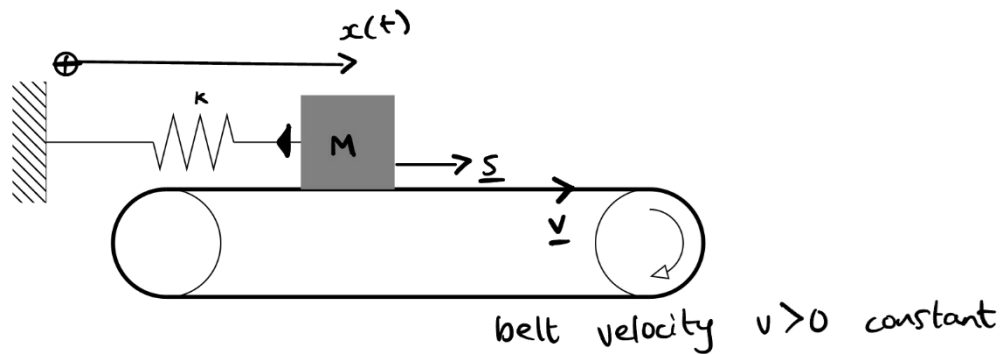
The report proceeds as follows. First, the governing equations for both the stick and slip phases are derived and the key assumptions stated. Numerical solutions are then computed to explore how the amplitude and period of the motion depend on the system parameters, particularly the mass and belt velocity. An analytical treatment of the slip-phase dynamics is compared with the numerical results. Finally, the model's limitations are critically assessed and possible extensions discussed.

Model Description:

The system under consideration consists of a block of mass M connected to a fixed wall by a linear spring of stiffness k , resting on a belt that moves at constant velocity v (Figure 1). This setup is directly analogous to the interaction between a bow and a string: the belt plays the role of the bow providing a continuous energy input, while the mass–spring system represents the oscillating string with its elastic restoring

force. The displacement of the block from its natural length position is denoted $x(t)$, and a dot denotes differentiation with respect to time.

Figure 1



Parameters and Variables

The model is governed by the following parameters: block mass M (kg), spring stiffness k (N/m), belt velocity v (m/s), static friction coefficient μ_s , and dynamic friction coefficient μ_d , with $\mu_s > \mu_d$. The normal reaction is $R = Mg$, where $g = 9.81\text{m/s}^2$ is gravitational acceleration. The key output variables are the displacement $x(t)$, velocity $\dot{x}(t)$, and the amplitude A and period T of the resulting oscillations.

The relative velocity between the block and belt is defined as

$$v_{\text{rel}} = \dot{x} - v.$$

This quantity determines whether the system is in the stick or slip phase and is central to the friction model.

Friction Model

In the simplest formulation, friction switches discretely between two constant values: a static friction force $S = \mu_s R$ during sticking, and a dynamic friction force $S = \mu_d R$ during slipping. After researching different approaches to modelling the friction discontinuity, the switch model framework presented by Leine, Van Campen, and De Kraker (1998) was identified as particularly well-suited to this problem. This avoids the calculation of smoothing methods, which approximate the discontinuous friction curve with a continuous function but produce difficult differential equations and integrations.

The switch model distinguishes three modes based on the relative velocity v_{rel} and a small numerical tolerance $\eta \ll v$:

Stick phase ($|v_{\text{rel}}| < \eta$ and $|kx| \leq \mu_s R$): The block moves with the belt, so $\dot{x} = v$ and the spring stretches linearly with time. The spring force $T = kx$ increases until it reaches the critical breakaway value $T^* = \mu_s R$, at which point the static friction can no longer hold the block. The critical displacement at which slipping begins is therefore

$$x^* = \frac{\mu_s R}{k} = \frac{\mu_s M g}{k}.$$

Stick-to-slip transition ($|v_{\text{rel}}| < \eta$ and $|kx| > \mu_s R$): The applied spring force exceeds the maximum static friction, so the block begins to accelerate. The friction force is set to $\mu_s R \operatorname{sgn}(kx)$ during this brief transition.

Slip phase ($|v_{\text{rel}}| > \eta$): Once sliding, the friction drops to the dynamic value $S = \mu_d R$, acting in the direction opposing relative motion. Newton's second law gives the equation of motion:

$$M\ddot{x} = -kx + \mu_d M g.$$

This is a second-order linear ODE describing simple harmonic motion about a shifted equilibrium position $x_{\text{eq}} = \mu_d M g / k$, with natural frequency

$$\omega = \sqrt{\frac{k}{M}}$$

The general solution during the slip phase can be written as

$$x(t) = x_{\text{eq}} + C_1 \cos(\omega t) + C_2 \sin(\omega t),$$

where C_1 and C_2 are determined by the displacement and velocity at the moment of transition from stick to slip.

Slip-to-stick transition: The block continues to oscillate until its velocity matches the belt velocity again ($\dot{x} = v$), at which point the block re-engages with the belt and the cycle repeats.

Approximations and Simplifications

Several simplifying assumptions underpin this model. The friction coefficients μ_s and μ_d are treated as constants, independent of velocity, contact time, or temperature. In reality, the kinetic friction force often depends on sliding speed (Leine et al. model) this velocity dependence using a curve of the form $F(v_{\text{rel}}) = -F_s \operatorname{sgn}(v_{\text{rel}}) / (1 + \delta |v_{\text{rel}}|)$, where the parameter δ controls how rapidly friction decays with increasing relative velocity. The present model neglects this effect for simplicity. Additionally, the belt is assumed to move at perfectly constant velocity with no compliance or vibration of its own, the spring is assumed to be massless and perfectly linear, and the motion is restricted to one dimension. The finite stick-band tolerance η is a numerical device required because a digital solver will rarely compute an exact relative velocity of zero; it does not represent a physical parameter but must be chosen small enough (typically $\eta \sim 10^{-6}$) not to affect results.

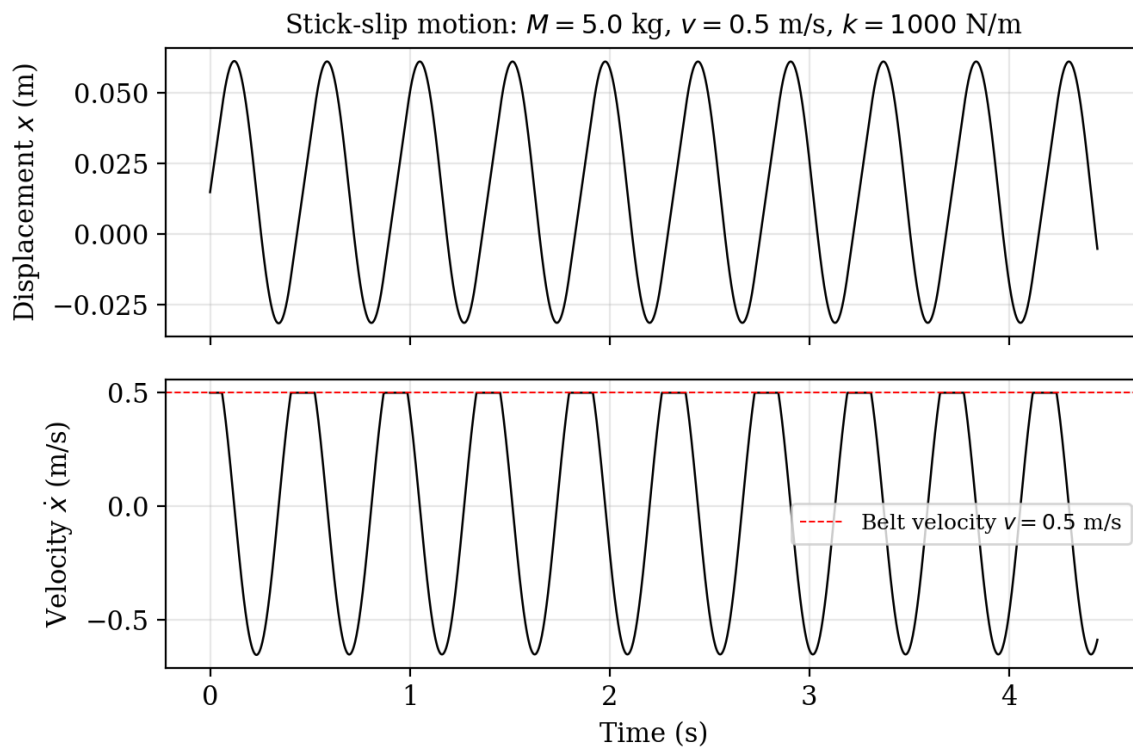
Model Output

The model produces time histories of $x(t)$ and $\dot{x}(t)$, from which the steady-state amplitude A and period T of the stick-slip oscillation can be extracted. The numerical implementation uses MATLAB's ODE solvers to integrate the switch model equations, with the phase logic checked at each timestep to determine which set of equations to apply. The analytical solution for the slip phase provides a means of independently verifying the numerical results and understanding how A and ω depend on the system parameters.

Results:

The model was implemented numerically using the switch model framework with parameter values $k = 1000\text{N/m}$, $\mu_s = 0.9$, $\mu_d = 0.3$, and $g = 9.81\text{m/s}^2$. Figure 2 shows the displacement and velocity time histories for a baseline case of $M = 5\text{kg}$ and $v = 0.5\text{m/s}$. The characteristic stick-slip behaviour is clearly visible: during the stick phase, the displacement increases linearly as the block moves with the belt at $\dot{x} = v$, while during the slip phase the block oscillates about the shifted equilibrium $x_{\text{eq}} = \mu_d Mg/k$, with velocity dropping well below zero before returning to match the belt. The motion settles into a steady periodic cycle after approximately one period.

Figure 2



To compare with analytical predictions, the slip-phase ODE $M\ddot{x} = -kx + \mu_d Mg$ was solved. At the moment of breakaway, $x = x^* = \frac{\mu_s Mg}{k}$ and $\dot{x} = v$, giving the slip-phase amplitude as $A_{\text{slip}} = \sqrt{(x^* - x_{\text{eq}})^2 + (v/\omega)^2}$, where $\omega = \sqrt{\frac{k}{M}}$ is the natural frequency.

Figure 3 shows how amplitude A and period T vary with mass M at fixed belt velocity $v = 0.5\text{m/s}$. Both increase monotonically with M : heavier blocks stretch the spring further before breakaway and oscillate more slowly. The numerical amplitude agrees very closely with the analytical slip-phase prediction across the full range $1 \leq M \leq 10\text{kg}$. For the period, the numerical values consistently exceed the analytical slip-only estimate of $\frac{2\pi}{\omega}$, and the divergence grows with M . This is expected, the full period includes a stick phase whose duration increases with mass (since a heavier block takes longer to reach the breakaway displacement at fixed belt speed), whereas $\frac{2\pi}{\omega}$ accounts only for the slip oscillation.

Figure 3

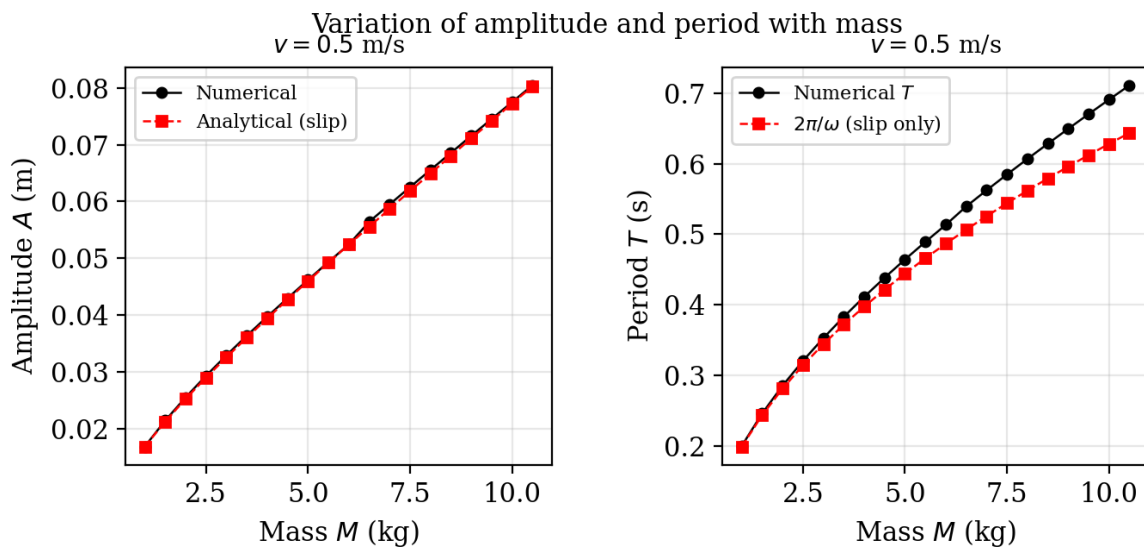


Figure 4 shows the effect of varying belt velocity v at fixed $M = 5\text{kg}$. Amplitude increases approximately linearly with v , again matching the analytical prediction well. The period behaviour is more revealing: at low belt speeds the total period is significantly longer than $\frac{2\pi}{\omega}$ because the stick phase dominates, the block must travel a large distance at a slow speed to reach breakaway. As v increases, the stick phase shortens and the total period converges toward the slip-only value of $2\pi/\omega \approx 0.44\text{s}$. In the bowed-string similarity, this corresponds to the observation that a faster bow speed produces louder sound (larger amplitude) at nearly the same pitch (frequency governed by string properties, not bow speed).

Critical Analysis

The results produced by the model are reasonable. The displacement and velocity traces exhibit the characteristic sawtooth-like periodicity expected of stick-slip systems, with the block accelerating during stick, overshooting during slip, and recapturing at the belt velocity. The trend with mass and belt velocity shows a heavier block stores more elastic energy before breakaway, producing larger amplitude oscillations at a lower frequency, while a faster belt increases the kinetic energy per cycle, raising the amplitude without significantly changing the natural frequency $\omega = \sqrt{k/M}$, which depends only on the spring and mass.

Furthermore, the slip-phase dynamics are well captured by the second-order ODE $M\ddot{x} = -kx + \mu_d Mg$. The difference in the period comes entirely from the stick phase, which isn't captured in the analytical solution. This is most pronounced at low belt velocities, where the block must travel slowly across the full stick displacement, making the stick phase a large fraction of the total cycle. In the context of a bowed string, this mirrors the physical reality that bow speed primarily controls loudness (amplitude) while pitch (frequency) is governed by the string's own mass and tension.

However, several simplifications limit the model's fidelity. The most significant is the assumption of constant friction coefficients. In practice, kinetic friction typically decreases with increasing sliding speed — Leine et al. model this with the curve $F(v_{\text{rel}}) = -F_s \operatorname{sgn}(v_{\text{rel}})/(1 + \delta |v_{\text{rel}}|)$, where the parameter δ controls the rate of decay. A velocity-dependent friction model would alter the shape of the slip-phase oscillation and could change the conditions under which stick-slip motion is sustained. A second limitation is the treatment of static friction as instantaneous: in reality, the breakaway force increases with the duration of contact, an effect Leine et al. capture through a time-dependent static friction model where F_{stick} grows exponentially toward a maximum during the stick phase. Including this would shift the breakaway point and break the symmetry of the limit cycle.

The one-dimensional, single degree of freedom result also constrains the model. A real bowed string undergoes both lateral and torsional vibration, as a bow drags the string sideways, it also rolls. Leine et al. demonstrated that for thicker strings the torsional mode has a much higher natural frequency than the lateral mode, so the 1-DOF approximation works well. But for thinner strings (where the torsional frequency drops closer to the lateral frequency), the two modes couple and the motion can become almost periodic or even chaotic, fundamentally changing their tone. The present model cannot capture this behaviour.

Conclusions

To summarise the mass–spring–belt model was developed to investigate stick-slip vibrations, motivated by my own musical interest in the bow–string interaction that produces sustained tones in stringed instruments. Using a switch model framework, the system was simulated numerically across a range of masses ($1 \leq M \leq 10$ kg) and belt velocities ($0.1 \leq v \leq 1$ m/s). The model successfully reproduces periodic stick-slip motion with physically consistent behaviour: amplitude increases with both M and v , while the total period increases with M and decreases with v , converging toward the slip-only period $\frac{2\pi}{\omega}$ at high belt speeds. The predictions for the slip-phase amplitude agree closely with numerical results, illustrating that the simplified ODE describes the essential oscillation dynamics. The main limitation is the constant-friction assumption; incorporating velocity-dependent or time-dependent friction, and extending to multiple degrees of freedom, would improve the model's ability to represent real systems such as bowed strings, where friction characteristics play important roles in determining the quality of tone.